

ADVANCED LINEAR ALGEBRA

HOMEWORK 6

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P.4.9. If $A \in \mathbf{M}_n$ is to be represented as $A = XY^\top$ for some $X, Y \in \mathbf{M}_{n \times r}$, explain why r cannot be smaller than $\text{rank}(A)$.

Solution. □

P.4.10. For $X \in \mathbf{M}_{m \times n}$, $\nu(X)$ denote the nullity of X .

(a) Show that $\nu(X^\top) = \nu(X) + m - n$.

(b) Let $A \in \mathbf{M}_{m \times k}$

Solution. □

P.4.13.

Solution. □

P.4.35.

Solution. □

P.4.38.

Solution. □